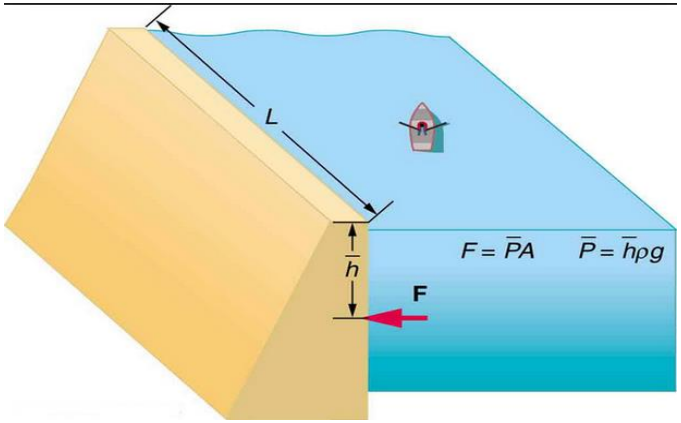
Kwame Nkrumah University of
Science & Technology, Kumasi, Ghana

Introduction to Fluid Mechanics
ME 251

Pressure in Fluids

David A. Quansah, PhD
Y. A. K. Fiagbe, PhD
Department of Mechanical Engineering, KNUST




L
$\bar{h}$
F
$F = \bar{P}A$
$\bar{P} = \bar{h}\rho g$

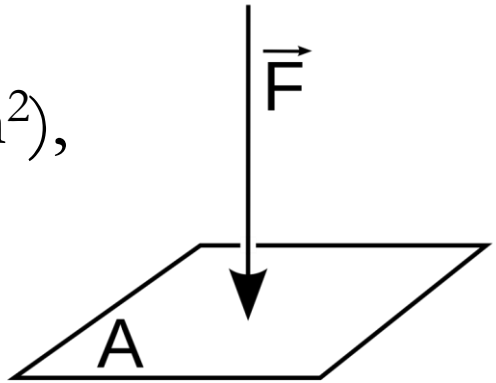
Areas to Cover

- Pressure in Fluid
  - ✓ Pressure variation in a fluid at rest
  - ✓ Absolute, gauge, atmospheric and vacuum pressures
  - ✓ Measurement of pressure: Manometers, mechanical gauges

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PRESSURE

- Pressure is the normal force per unit area.
- Its unit is **Newton per square meter** (N/m^2), or Pascal (Pa).

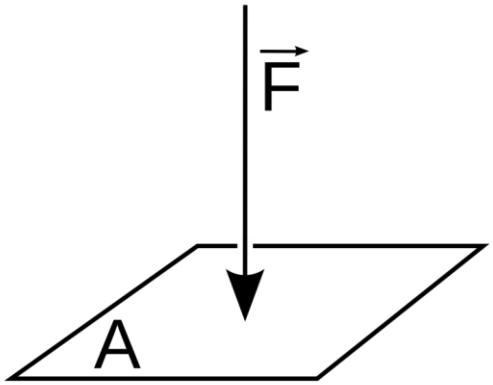


- The unit **Pascal** is too small for pressures encountered in practice.
  - ✓ Multiples of the Pascal are often used e.g. kilopascal (1 kPa $=10^3$ Pa) and the megapascal (1 MPa = 10^6 Pa).

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PRESSURE

- Three other pressure units commonly used in practice, are:
  - ✓ **bar,**
  - ✓ **standard atmosphere, and**
  - ✓ **kilogram-force per square centimeter**



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PRESSURE

- English system (non-SI):

  ✓ the unit for pressure is the pound-force per square inch (lbf/in^2,or psi),

  ✓ 1 atm = 14.696 psi.

  ✓ The pressure units kgf/cm^2 and lbf/in^2 are also denoted by kg/cm^2 and lb/in^2, respectively,

Some useful unit conversions

$$1\ \text{bar} = 10^5\ \text{Pa} = 0.1\ \text{MPa} = 100\ \text{kPa}$$

$$1\ \text{atm} = 101{,}325\ \text{Pa} = 101.325\ \text{kPa} = 1.01325\ \text{bars}$$

$$1\ \text{kgf/cm}^2 = 9.807\ \text{N/cm}^2 = 9.807 \times 10^4\ \text{N/m}^2 = 9.807 \times 10^4\ \text{Pa}$$

$$= 0.9807\ \text{bar}$$

$$= 0.9679\ \text{atm}$$

PRESSURE

- The actual pressure at a given position is **absolute pressure**, measured relative to absolute vacuum (i.e., absolute zero pressure).

- Most pressure measuring devices are calibrated to read zero in the atmosphere (neglect atmospheric pressure).

- They measure the difference between the absolute pressure and the local atmospheric pressure. This difference is called the **gauge pressure**.

- Pressures below atmospheric pressure are called **vacuum pressures.** They are measured by vacuum gauges that indicate the difference between the atmospheric pressure and the absolute pressure.

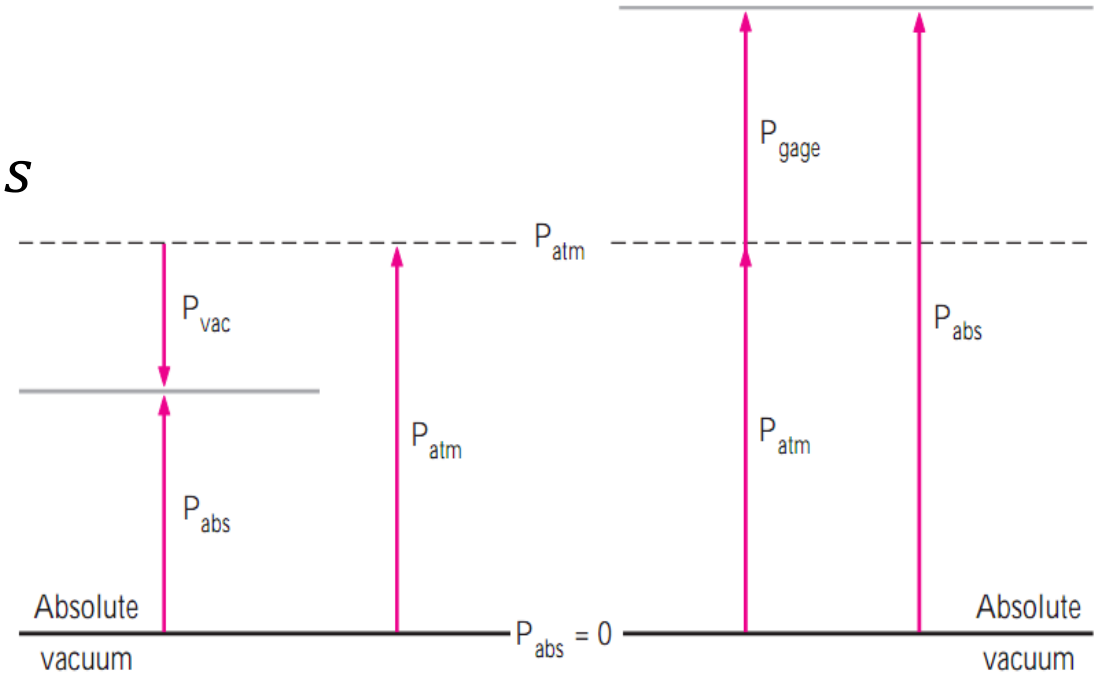
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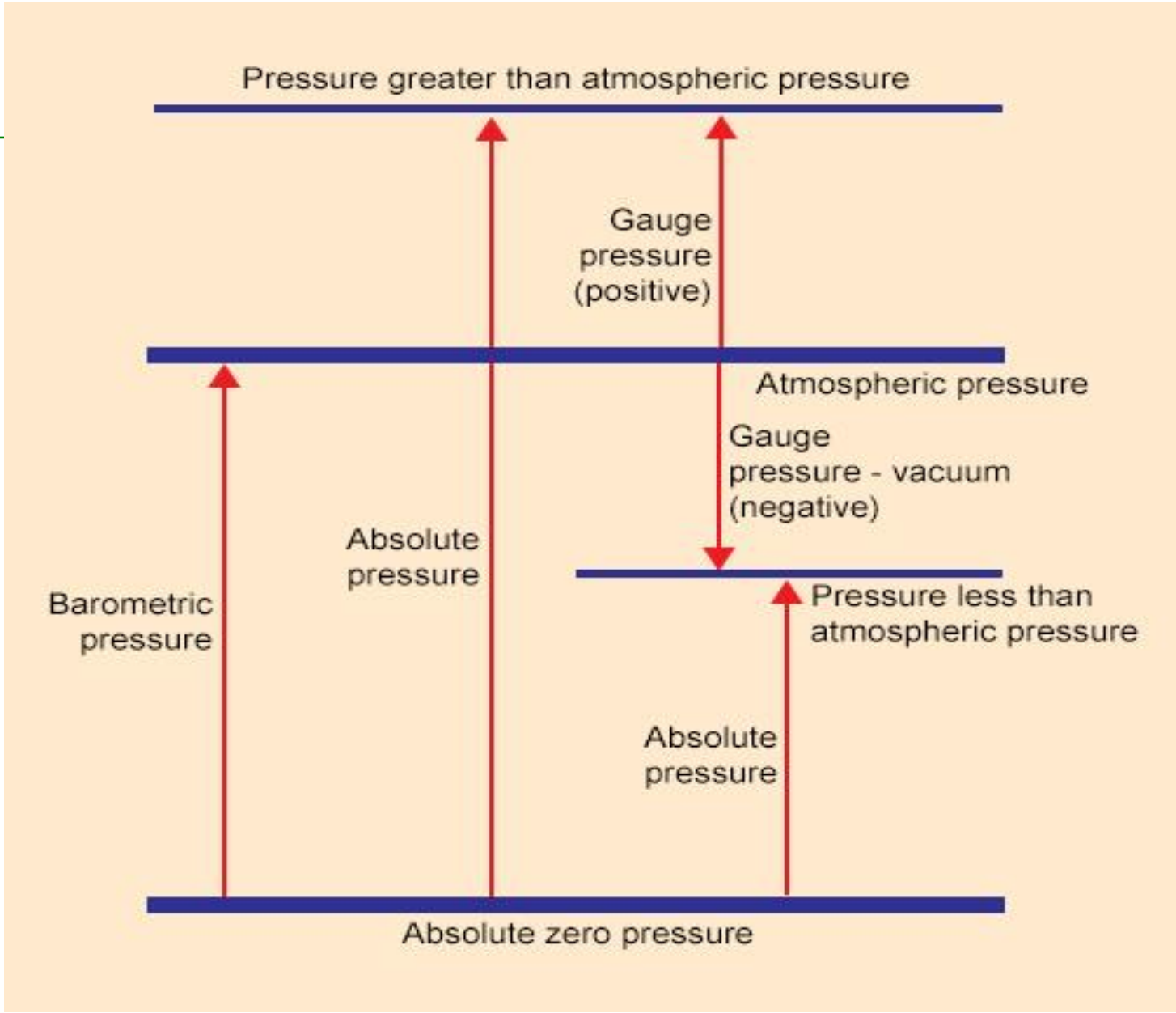
PRESSURE

- Absolute, gauge, and vacuum pressures are all positive quantities and are related to each other by as follows:

$$P_{gauge} = P_{abs} - P_{atm}$$

$$P_{vac} = P_{atm} - P_{abs}$$




Pressure greater than atmospheric pressure
Gauge pressure (positive)
Atmospheric pressure
Gauge pressure - vacuum (negative)
Absolute pressure
Barometric pressure
Pressure less than atmospheric pressure
Absolute pressure
Absolute zero pressure

Question:

A vacuum gauge connected to a chamber reads 5.8 psi at a location where the atmospheric pressure is 14.5 psi. Determine the absolute pressure in the chamber.

Recall:

$$P_{vac} = P_{atm} - P_{abs}$$

The gauge pressure of the chamber is given as 5.8 psi (Vacuum Pressure).

Atmospheric pressure is given as 14.5 psi

Absolute pressure is therefore:

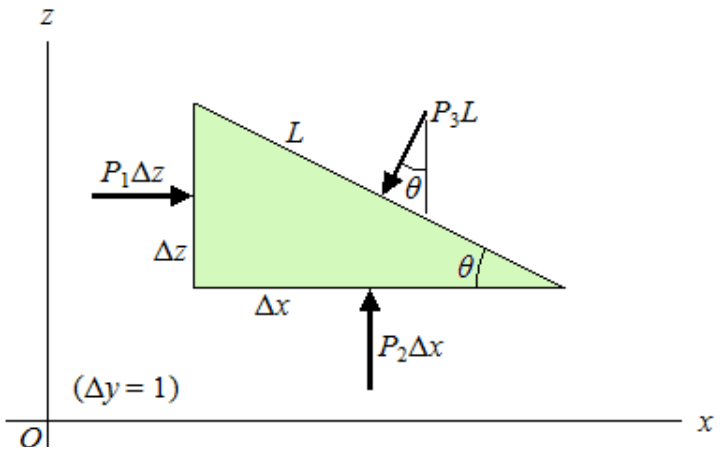
$$P_{abs} = P_{atm} - P_{vac}$$

$$P_{abs} = 14.5 - 5.8$$
$$= 8.7 \text{ psi}$$

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Pressure at a Point

- The mean pressures at the three surfaces are P_1, P_2,and P_3
- The force acting on a surface is the product of mean pressure and the Area



$$\sum F_x = ma_x = 0: \qquad P_1\,\Delta z - P_3 l \sin\theta = 0$$

$$\sum F_z = ma_z = 0: \qquad P_2\,\Delta x - P_3 l \cos\theta - \frac{1}{2}\rho g\,\Delta x\,\Delta z = 0$$

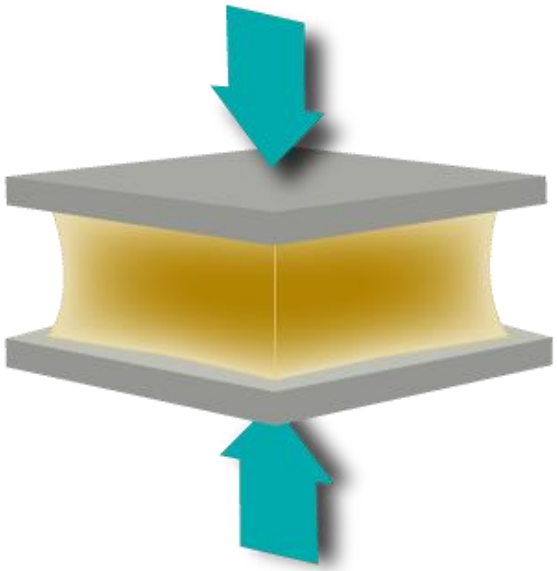
$$P_1 - P_3 = 0$$

$$P_2 - P_3 - \frac{1}{2}\rho g\,\Delta z = 0$$

$$P_1 = P_2 = P_3 = P$$

Pressure at a Point

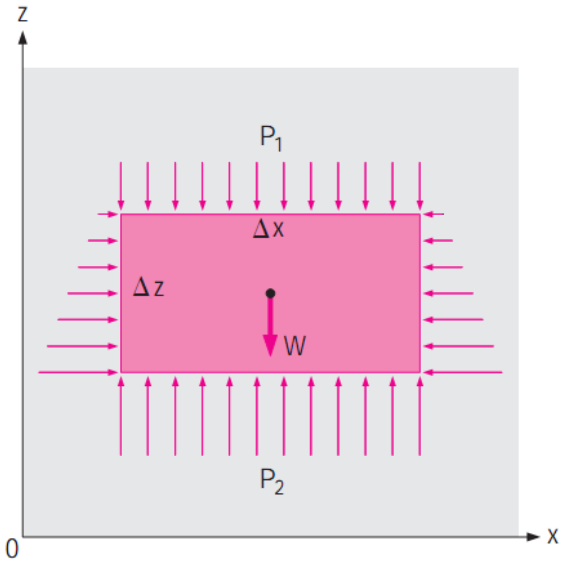
- Pressure is the compressive force per unit area, and it gives the impression of being a vector.
- Pressure at any point in a fluid is the same in all directions.
- That is, it has magnitude but not a specific direction.
  - **It is a scalar quantity**.



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Variation of Pressure with Depth

- Consider a rectangular fluid element of height Δz, length Δx, and unit width in equilibrium.
- Density of the fluid ρ is constant
- A force balance in the vertical z-direction gives



$$\sum F_z = ma_z = 0: \qquad P_2\,\Delta x - P_1\,\Delta x - \rho g\,\Delta x\,\Delta z = 0$$

$$\Delta P = P_2 - P_1 = \rho g\,\Delta z = \gamma_s\,\Delta z$$

The pressure difference between two points in a constant density fluid is proportional to the vertical distance **Δz** between the points and the density ρ of the fluid. Pressure in a fluid increases linearly with depth.

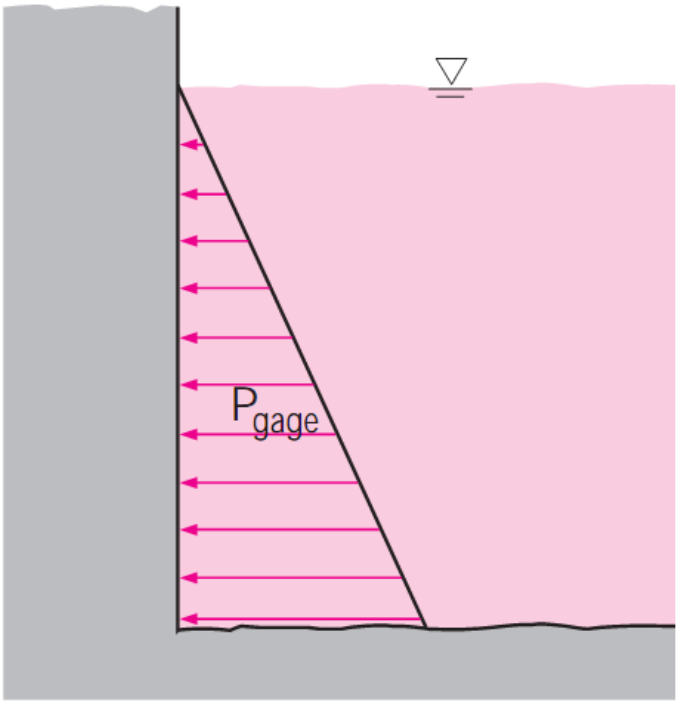
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Variation of Pressure with Depth

- For a fluid whose density changes significantly with elevation, a relation for the variation of pressure with elevation can be obtained by dividing by $\Delta x \Delta z$, and taking the limit as $\Delta z \rightarrow 0$.

$$P_2 \Delta x - P_1 \Delta x - \rho g \Delta x \Delta z = 0$$

$$\frac{dP}{dz} = -\rho g$$

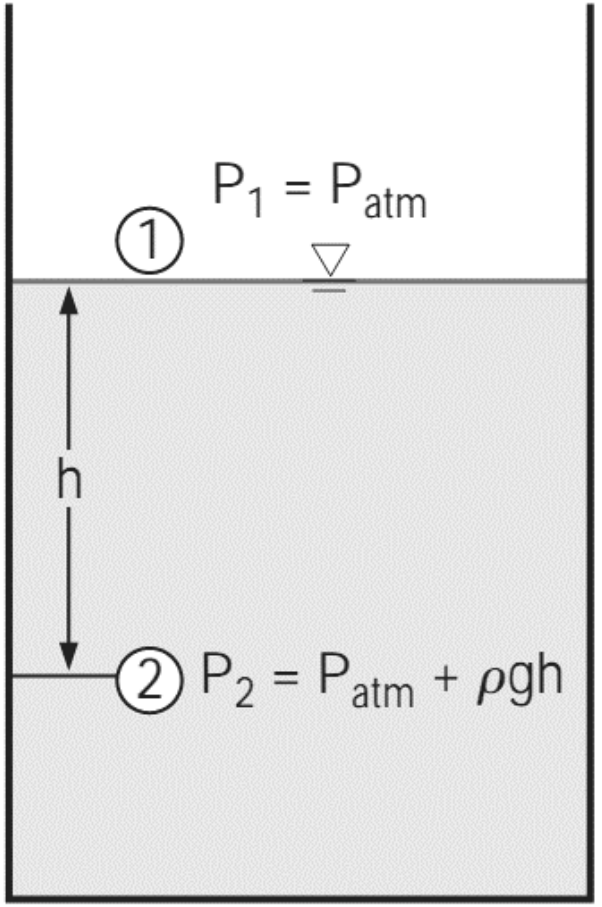


Variation of Pressure with Depth

- The pressure difference between points 1 and 2 can be determined by integrating

$$\frac{dP}{dz} = -\rho g$$

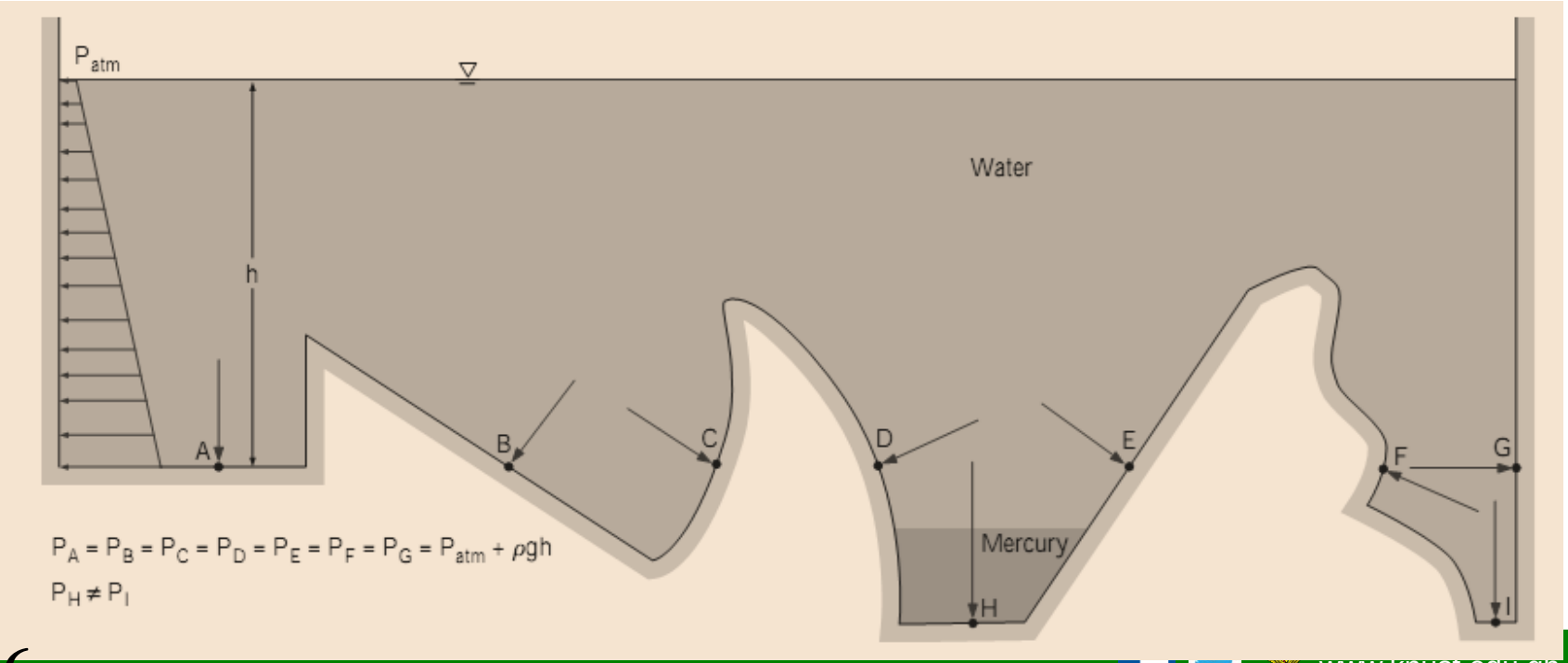
$$\Delta P = P_2 - P_1 = -\int_1^2 \rho g \, dz$$



Pressure in a fluid at rest

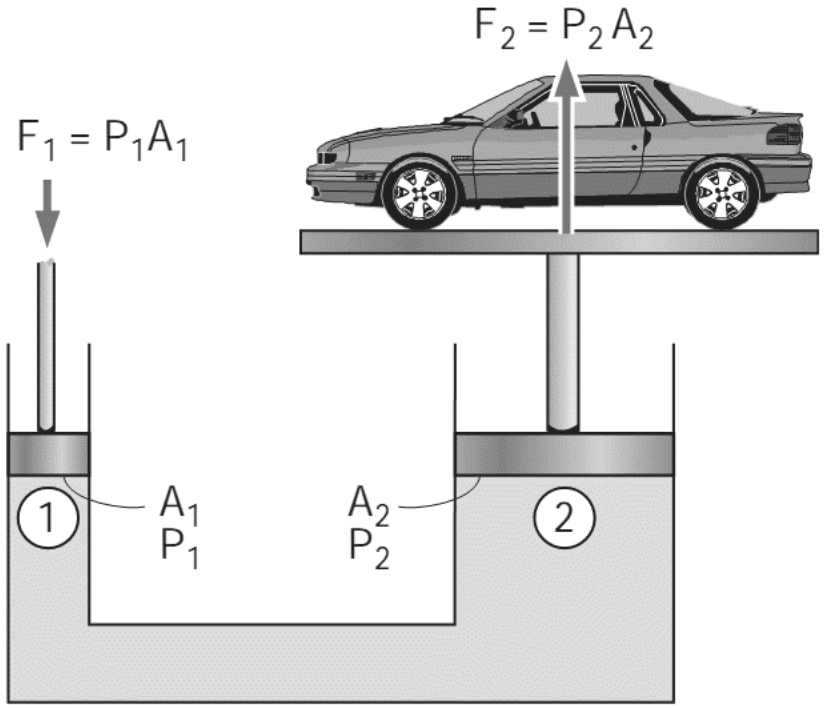
- Pressure change is independent of the shape or cross section of the container.

- It changes with the vertical distance, but remains constant in Horizontal direction.



Pascal's Law

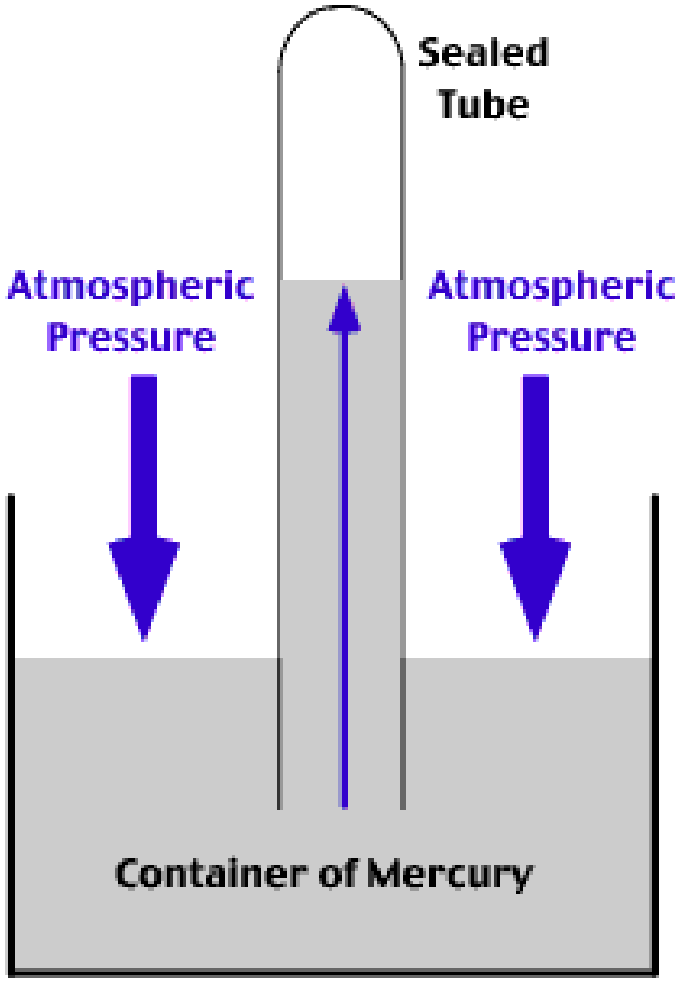
- The pressure applied to a confined fluid is transmitted by the same amount.
- A consequence of the pressure in a fluid remaining constant in the horizontal direction,



$$P_1 = P_2 \quad \rightarrow \quad \frac{F_1}{A_1} = \frac{F_2}{A_2} \quad \rightarrow \quad \frac{F_2}{F_1} = \frac{A_2}{A_1}$$

THE BAROMETER

- A device for measuring atmospheric pressure.
- It is Inverted mercury-filled tube in a mercury container that is open to the atmosphere.

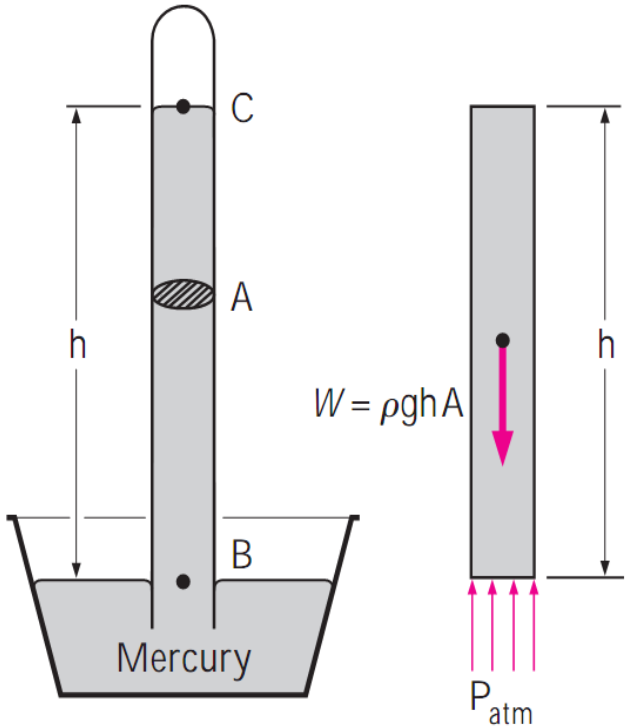


THE BAROMETER

- The pressure at point B is equal to the atmospheric pressure.
-
- Pressure at C can be taken to be zero since there is only mercury vapor above point C and the pressure is very low relative to P_{atm} and can be neglected to an excellent approximation.

- Writing a force balance in the vertical direction gives:

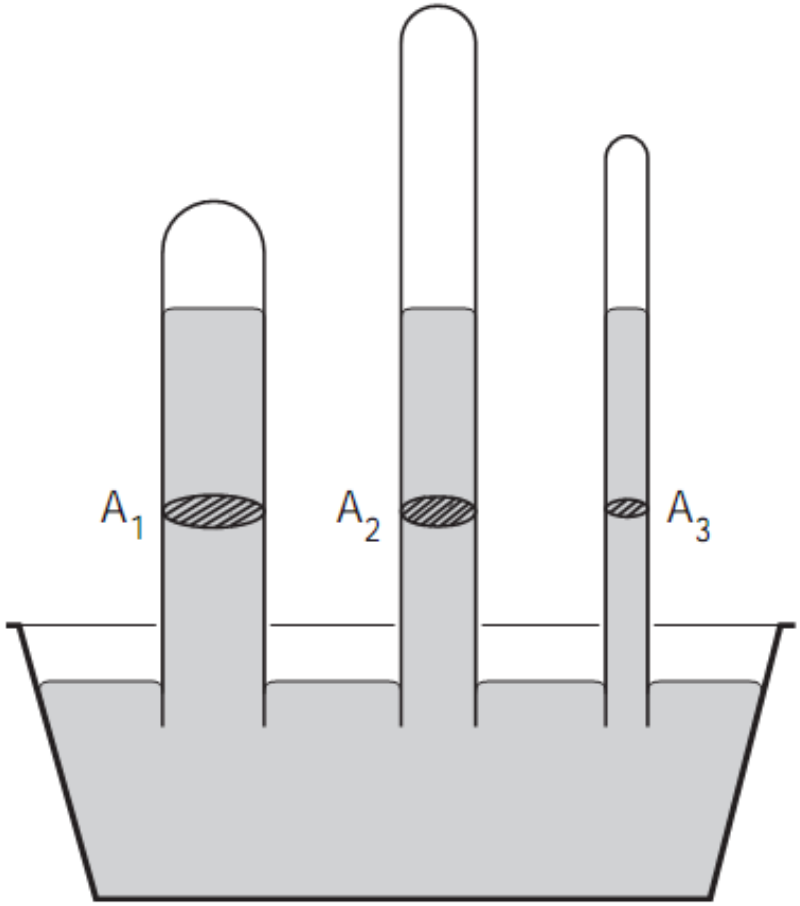
$$P_{atm} = \rho g h$$



Atmospheric pressure = Barometric pressure

THE BAROMETER

- Effect of cross-sectional area is negligible



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Atmospheric Pressure Unit

- A frequently used pressure unit is the standard atmosphere.
  - ✓ The pressure produced by a column of mercury 760 mm in height at 0°C (ρ_{Hg} =13,595 kg/m^3) under standard gravitational acceleration (g = 9.807 m/s^2).
- If water instead of mercury were used to measure the standard atmospheric pressure, a water column of about 10.3 m would be needed.
- Pressure is sometimes expressed (especially by weather forecasters) in terms of the height of the mercury column.
  - ✓ The standard atmospheric pressure, for example, is 760 mmHg (29.92 inHg) at 0°C.
  - ✓ The unit mmHg is also called the torr in honor of Torricelli.
  - ✓ Therefore, 1 atm = 760 torr and 1 torr = 133.3 Pa

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Question

Determine the atmospheric pressure at a location where the barometric reading is 740 mmHg and the gravitational acceleration is g = 9.81 m/s^2. Assume the temperature of mercury to be 10 °C, at which its density is 13,570 kg/m^3.

Applicable Equation:

$$P_{atm} = \rho . g . h$$

$$P_{atm} = (13570\ \text{kg/m}^3).(9.8\ m/s^2).(0.74\ m$$

$$= 98.5\ \text{kPa}$$

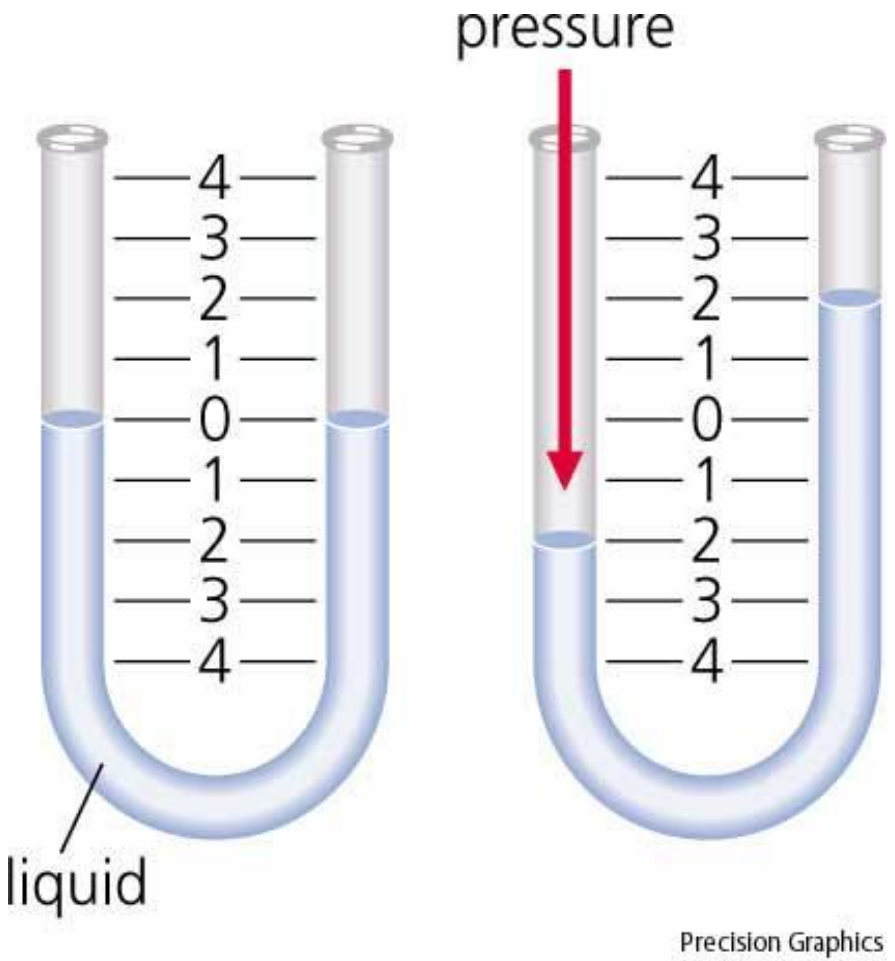
$Density\ of\ mercury\ at\ 10\ ^{o}\text{C}\ (\rho)$
$= 13570\ \text{kg/m}^3$

$Height\ of\ mercury\ column\ (h)$
$= 740\ mm = 0.74\ m$

$Acceleration\ due\ to\ gravity\ (g)$
$= 9.81\ \text{m/s}^2$

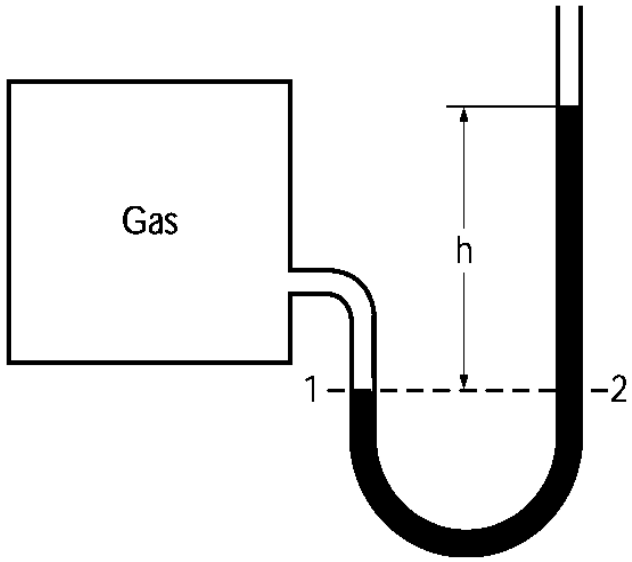
THE MANOMETER

- A manometer is a glass or plastic **U-tube** containing one or more fluids such as mercury, water, alcohol, or oil.
- They are commonly used to measure small and moderate pressure differences.



THE MANOMETER

- Since the gravitational effects of gases are negligible, the pressure anywhere in the tank and at position 1 have the same value.
- Since pressure in a fluid does not vary in the horizontal direction within a fluid, the pressure at point 2 is the same as the pressure at point 1, therefore $\mathbf{P_2 = P_1}$.
- The differential fluid column of height **h** is in static equilibrium, and it is open to the atmosphere.
- Pressure at point 2 is determined directly as

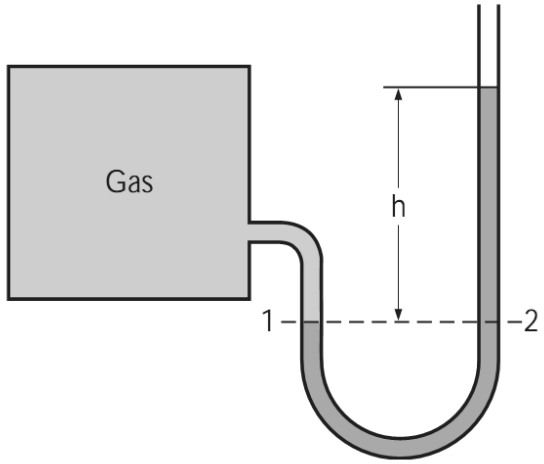


$$P_2 = P_{atm} + \rho gh$$

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THE MANOMETER

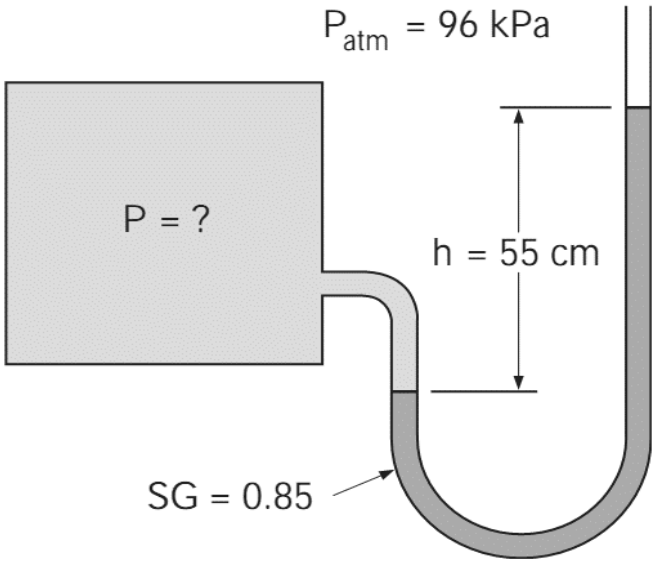
- The cross-sectional area of the tube has no effect on the differential height **h**, and thus the pressure exerted by the fluid.
-
- However, the diameter of the tube should be large enough (more than a few millimeters) to ensure that the surface tension effect and thus the capillary rise is negligible.



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Question

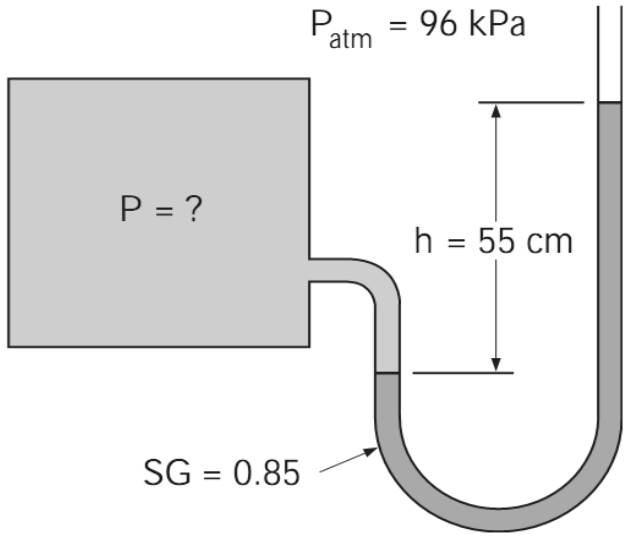
- A manometer is used to measure the pressure in a tank.
- The fluid used has a specific gravity of 0.85, and the manometer column height is 55 cm, as shown.
- If the local atmospheric pressure is 96 kPa, determine the absolute pressure within the tank.



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Questi

Applicable Equation:



$$P_{abs} = P_{atm} + P_{gauge}$$

$$P_{abs} = P_{atm} + \rho g h$$

$$\text{Density } (\rho) = 0.85 \times 1000 \frac{\text{kg}}{\text{m}^3}$$

$$= 850 \frac{\text{kg}}{\text{m}^3}$$

$$P_{abs} = 100.6 \text{ kPa}$$

Height of fluid column = 55 cm =0.55 m

Note:

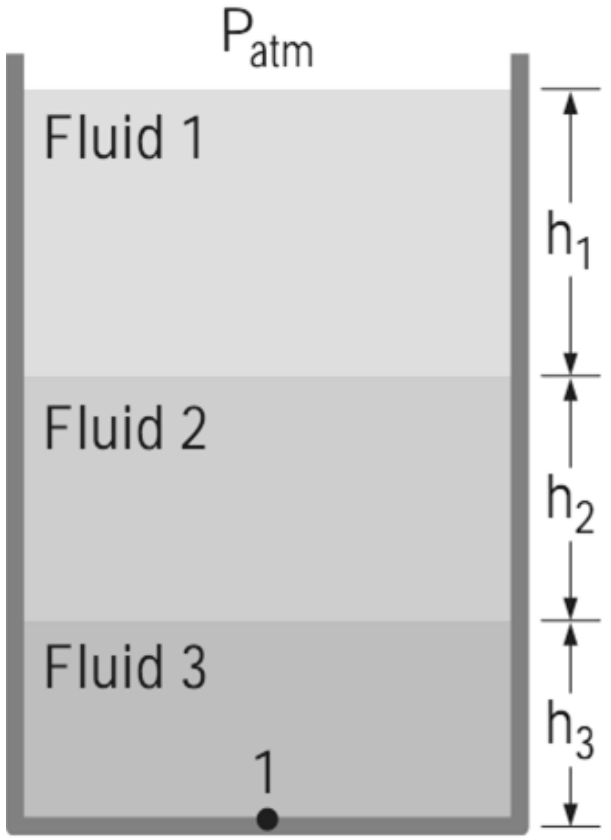
P_{gauge}= 4.6 kPa

P_{atm}=96 kPa =96000 Pa =96000 N/m^2

$$P_{abs} = 96 \text{ kPa} + \frac{(850 \text{ kg/m}^3)(9.8 \text{ m/s}^2)(0.55 \text{ m})}{1000}$$

Multifluid System

- The pressure change across a fluid column of height **h** is $\Delta P = \rho g h$,
- Pressure increases downward in a given fluid and decreases upward ($P_{bottom} > P_{top}$), and
- Two points at the same elevation in a continuous fluid at rest are at the same pressure.



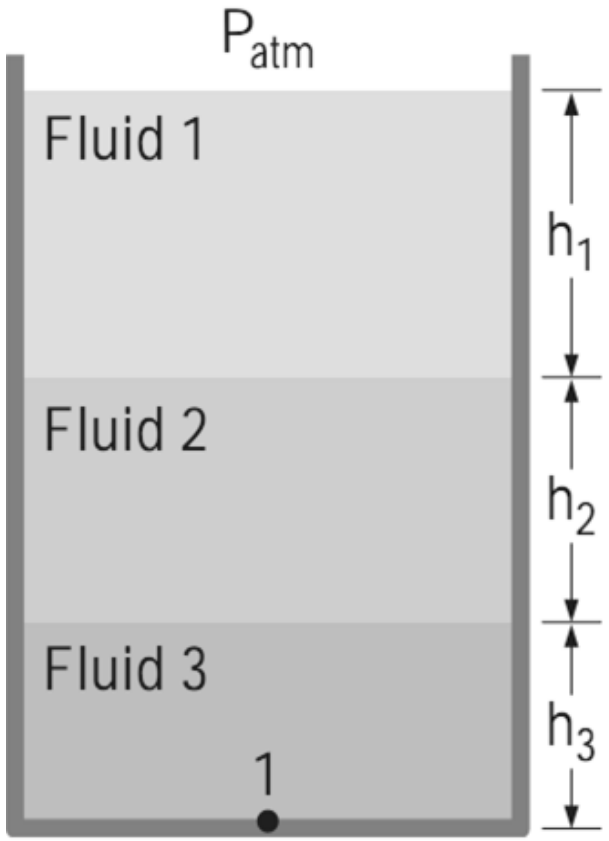
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Multifluid System

- The pressure at any point can be determined by starting with a point of known pressure and adding or subtracting

$$\rho gh$$

terms as we advance toward the point of interest.

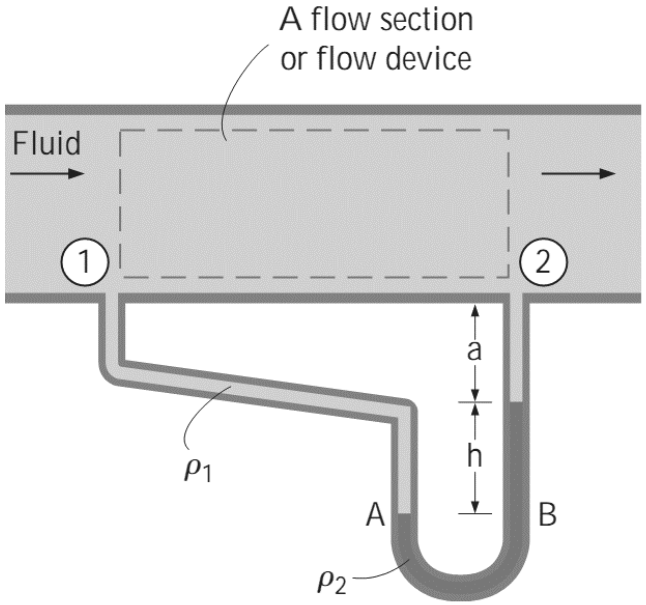


$$P_{atm} + \rho_1 g h_1 + \rho_2 g h_2 + \rho_3 g h_3 = P_1$$

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Pressure Difference

- The working fluid can be either a gas or a liquid with density, ρ_1.
- The density of the manometer fluid is ρ_2,
- The differential fluid height is **h**.
- Pressure difference $\mathbf{P_1 - P_2}$ is obtained by:
  - ✓ Moving along the tube, starting at point 1 with pressure P_1 and adding or subtracting the **ρgh** terms until we reach point 2, and setting the result equal to P_2.



$$P_1 + \rho_1 g(a + h) - \rho_2 gh - \rho_1 ga = P_2$$

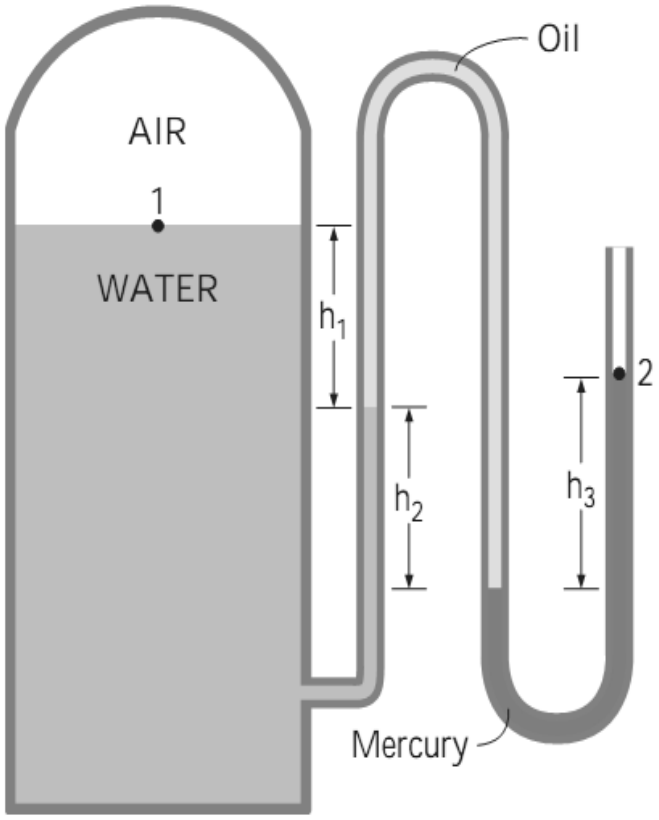
$$P_1 - P_2 = (\rho_2 - \rho_1)gh$$

Question

The water in a tank is pressurized by air, and the pressure is measured by a multifluid manometer as shown in Fig.
The tank is located on a mountain at an altitude of **1400 m** where the atmospheric pressure is **85.6 kPa**.

Determine the air pressure in the tank if $\mathbf{h_1 = 0.1\ m}$, $\mathbf{h_2 = 0.2\ m}$, and $\mathbf{h_3 = 0.35\ m}$.

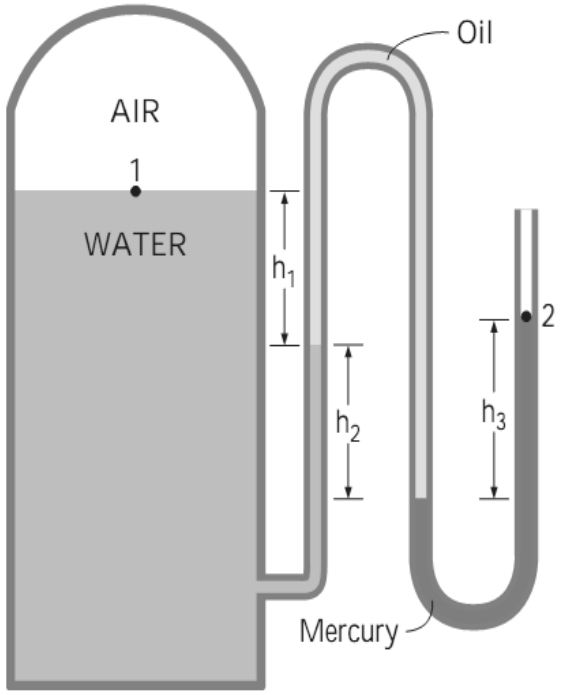
Take the densities of water, oil, and mercury to be **1000 kg/m³**, **850 kg/m³**, and **13,600 kg/m³**, respectively.



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Question

A general expression for the arrangement may be written as:



$$P_1 + \rho_{water} g h_1 + \rho_{oil} g h_2 - \rho_{mercury} g h_3 = P_2 = P_{atm}$$

$$P_1 = P_{atm} - \rho_{water} g h_1 - \rho_{oil} g h_2 + \rho_{mercury} g h_3$$

$$P_1 = P_{atm} + g\left(\rho_{mercury} h_3 - \rho_{water} h_1 - \rho_{oil} h_2\right)$$

$$P_1 = 85.6\ kPa + 9.81\ m/s^2 \frac{\left(13600\ \frac{kg}{m^3} \times 0.35\ m - \frac{1000\ kg}{m^3} \times 0.1\ m - \frac{850\ kg}{m^3} \times 0.2\ m\right)}{1000}$$

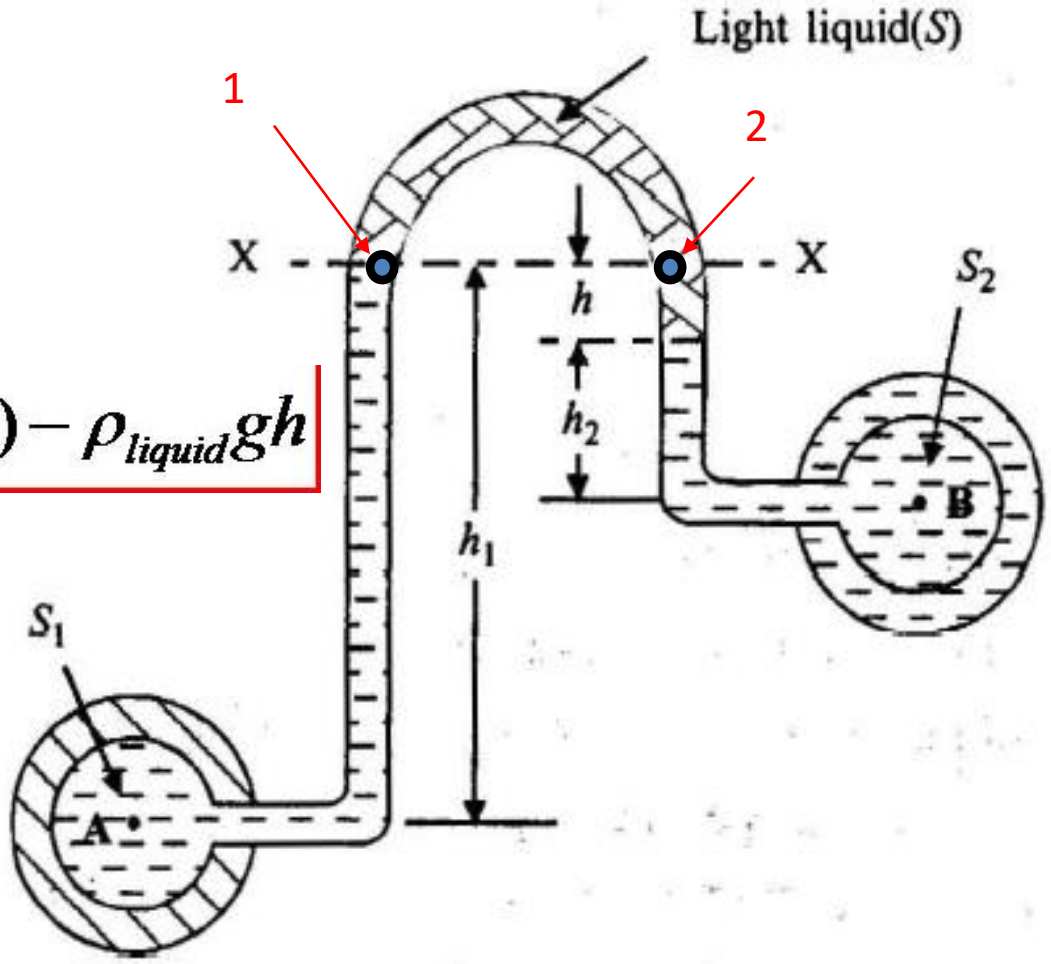
$$\boldsymbol{P_1 = 130\ \text{kPa}}$$

Inverted U-tube Differential Manometers

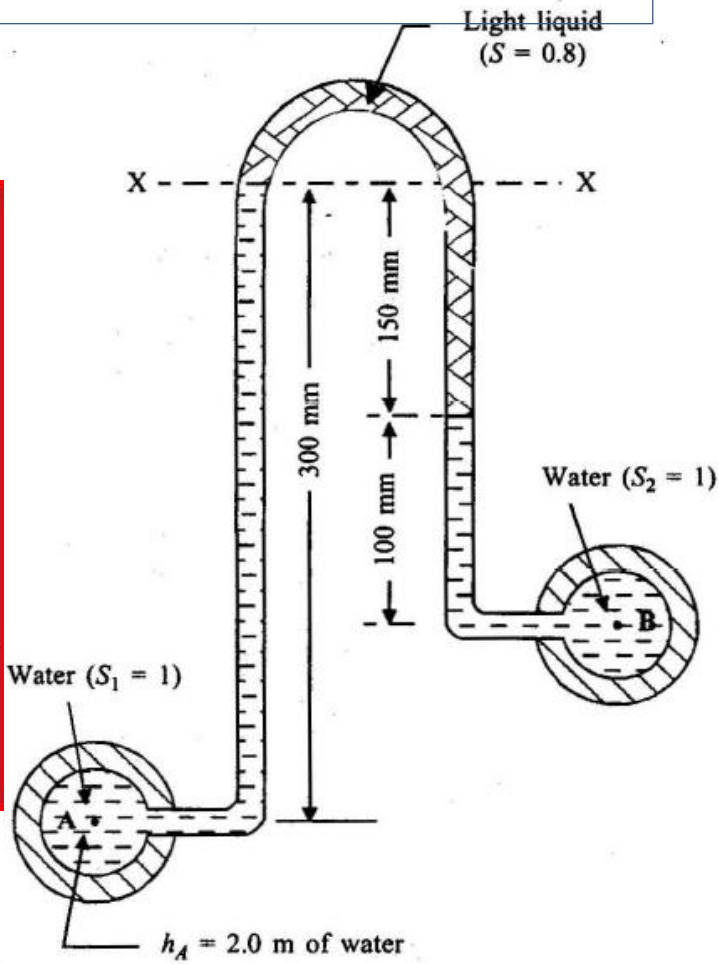
At x-x

$$P_1 = P_2$$

$$P_A - \rho_A g(h_1) = P_B - \rho_B g(h_2) - \rho_{liquid} gh$$



An inverted manometer having an oil of specific gravity 0.8 connected to two different pipes carrying water under pressure. Determine the pressure in the pipe B the pressure in the pipe A is 2.0 meters of water



At datum x - x

$P1 = P2$

$$P_A = 2 \times 1000 \times 9.81 = 19620 \text{ N/m}^2$$

$$P_A - \rho_A g(h_1) = P_B - \rho_B g(h_2) - \rho_{liquid} gh$$

$$19620 - 1000 \times 9.81 \times 0.3 = P_B - 1000 \times 9.81 \times 0.1 - 800 \times 9.81 \times 0.15$$

$$P_B = 18835 \text{ N/m}^2 = 18.8 \text{ kN/m}^2$$

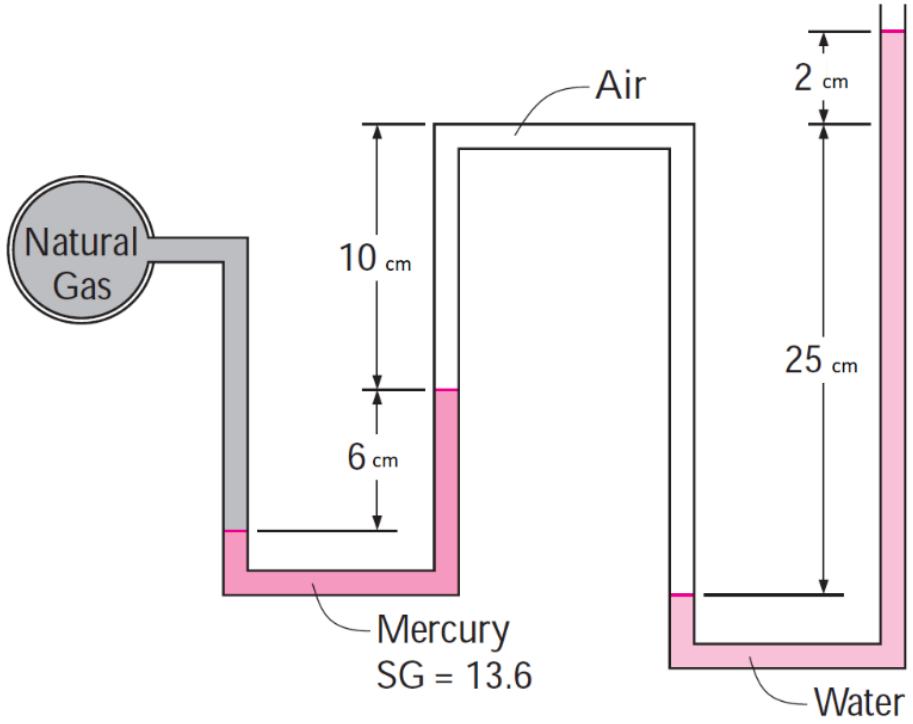
Useful Links

- On Manometry
  - https://www.youtube.com/watch?v=9SQ2FPHCZJk
- Bourdon Gauge
  - https://www.youtube.com/watch?v=-9IfpNAESlM

10 min (total)

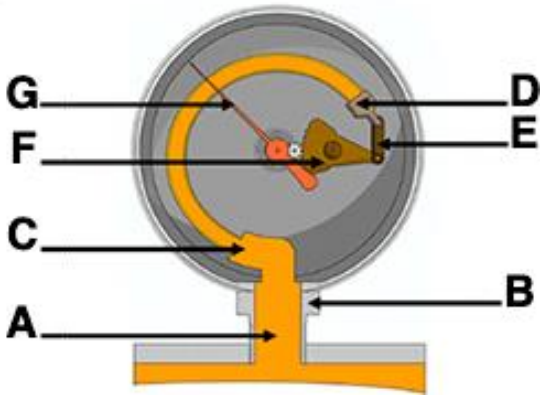
Question

The pressure in a natural gas pipeline is measured by the manometer shown in Figure 3 with one of the arms open to the atmosphere where the pressure is 760 mmHg. Determine the absolute pressure in the pipeline.



Other Pressure Measuring Devices

- This pressure gauge is attached to a pipeline at the inlet pipe (A).
- The inlet pipe is held in place by the socket block (B), which also holds the device to the process line.
- Pressure flows into the stationary end of the Bourdon tube (C).
- The pressure is distributed through the C-shaped Bourdon tube (D).
- As a result of this pressure, the C-shape straightens.
- A pivot and pivot pin (E) connect the straightening movement with the sector gear (F) at the moving end of the Bourdon tube.
- Due to the gearing system, the movement at the moving end of the Bourdon tube is amplified so that a very slight pressure change results in a considerable motion of the indicator needle (G).



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Pressure Transducers

- Pressure transducers use various techniques to convert the pressure effect to an electrical effect such as a change in voltage, resistance, or capacitance.

- They are smaller and faster, and can be more sensitive, reliable, and precise than their mechanical counterparts.
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- Pressures from less than a millionth of 1 atm to several thousands of atm.

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Pressure Transducers

- A wide variety available to measure gauge, absolute, and differential pressures in a wide range of applications.

  - **Gauge pressure** transducers use the atmospheric pressure as a reference

  - **Absolute pressure** transducers are calibrated to have a zero signal output at full vacuum.

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  - **Differential pressure** transducers measure the pressure difference.

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Other Pressure Transducers

- Strain-gauge Pressure Transducers
- Piezoelectric Transducers

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